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I Muscle Force Density in a 3D Musculoskeletal Region

I.1 Problem Statement

A biomechanics researcher analyzes a 3D musculoskeletal region where the **muscle force density** is given by:

$$g(x, y, z) = 652.75x^4y^8z^7$$

over the region bounded by: $0 \leq x \leq 5$, $0 \leq y \leq 6$, $0 \leq z \leq 3$. Compute the total muscle force distributed in the region.

I.2 Detailed Solution

Sol.

$$\begin{aligned}
F &= \iiint_V g(x, y, z)\, dV \\
&= \int_0^3 \int_0^6 \int_0^5 652.75x^4y^8z^7\, dx\, dy\, dz \\
&= 652.75 \left(\int_0^5 x^4\, dx\right) \left(\int_0^6 y^8\, dy\right) \left(\int_0^3 z^7\, dz\right) \\
&= 652.75 \left[\frac{x^5}{5}\right]_0^5 \left[\frac{y^9}{9}\right]_0^6 \left[\frac{z^8}{8}\right]_0^3 \\
&= 652.75\,(625)\,(1,119,744)\,(820.125) \\
&\approx \boxed{374,649,962,215,039.5000}
\end{aligned}$$

I.3 Matlab Implementation

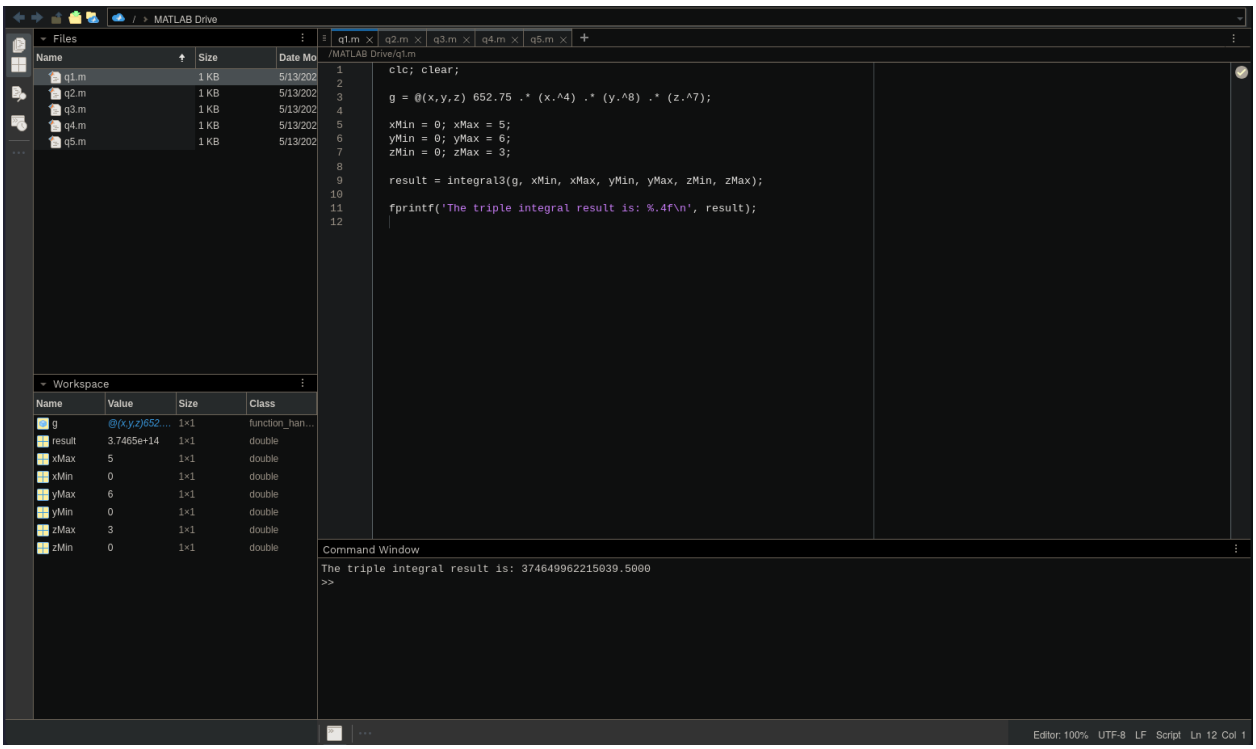

Figure I.1: Student 04 - Matlab Screenshot

I.4 Conclusion

The manual calculation for the total muscle force (Equation I.2) gives the **same** result as the MATLAB script (Figure I.1).

II Force Distribution in a Spherical Joint Model

II.1 Problem Statement

A biomechanics team models a spherical joint system with radius **7.5 units**, where the **joint force density** at a point (x, y, z) is described by:

$$I(x, y, z) = 600.17x^2 + 600.17y^2 + 600.17z^2$$

Compute the total joint force in the system using **spherical coordinates**.

II.2 Detailed Solution

Sol.

$$\begin{cases} I(\rho,\theta,\phi) &= 600.17(x^2+y^2+z^2) = 600.17\rho^2 \\ dV &= \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta \end{cases}$$

$$\begin{aligned} \Rightarrow F &= \int_0^{2\pi}\int_0^{\pi}\int_0^{7.5} (600.17\rho^2)(\rho^2\sin\phi)\, d\rho\, d\phi\, d\theta \\ &= 600.17\left(\int_0^{7.5}\rho^4\, d\rho\right)\left(\int_0^{\pi}\sin\phi\, d\phi\right)\left(\int_0^{2\pi} d\theta\right) \\ &= 600.17\left[\frac{\rho^5}{5}\right]_0^{7.5}\Big[-\cos\phi\Big]_0^{\pi}\Big[\theta\Big]_0^{2\pi} \\ &= 600.17\left(\frac{151875}{32}\right)(2)\,(2\pi) \\ &\approx \boxed{35,794,842.8185} \end{aligned}$$

II.3 Matlab Implementation

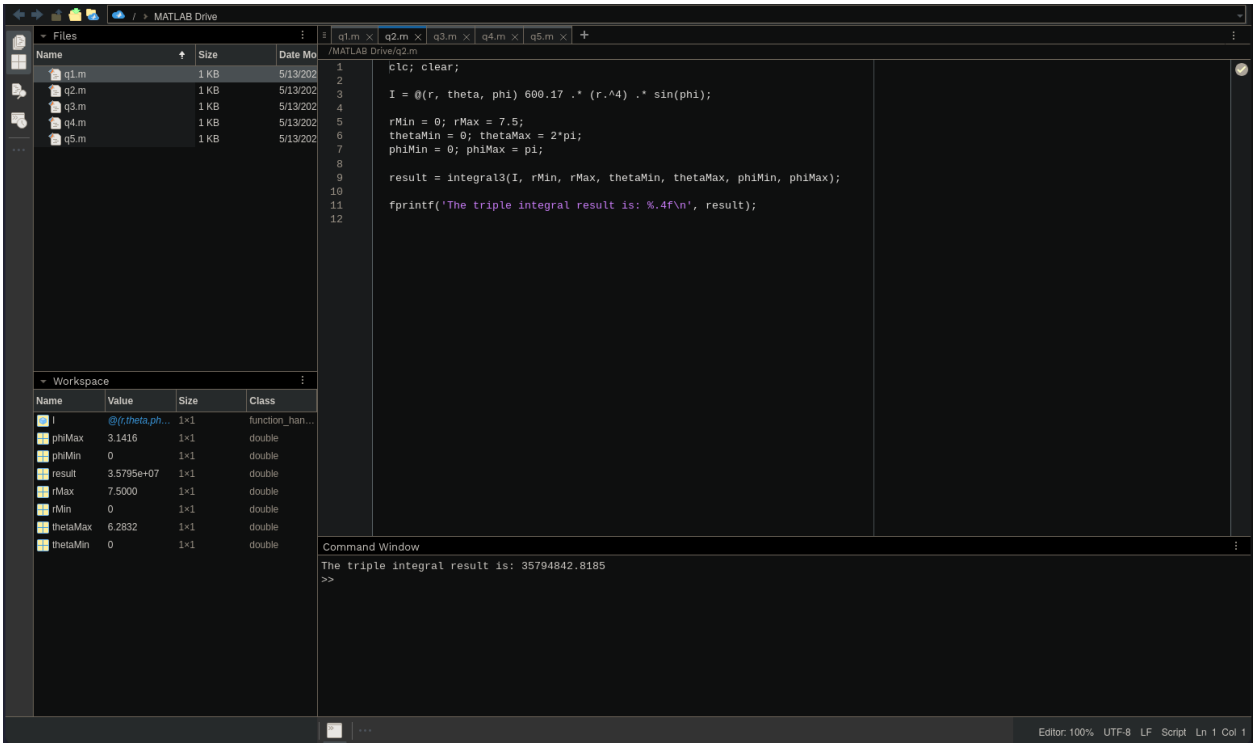

Figure II.1: Student 04 - Matlab Screenshot

II.4 Conclusion

By converting to spherical coordinates, the total joint force (Equation II.2) **does** match the MATLAB output (Figure II.1).

III Tendon Stress Distribution in a Cylindrical Limb Segment

III.1 Problem Statement

A cylindrical limb segment in a musculoskeletal model has a radius of **5.5 units** and a height of **9.5 units**. The **tendon stress density** at a point (x, y, z) within the segment is defined by:

$$P(x, y, z) = 354.62x^2 + 354.62y^2$$

Compute the total tendon stress in the segment using **cylindrical coordinates**.

III.2 Detailed Solution

Sol.

$$\begin{cases} P(r, \theta, z) & = 354.62(x^2 + y^2) = 354.62r^2 \\ dV & = r\,dr\,d\theta\,dz \end{cases}$$

$$\begin{aligned} \Rightarrow S &= \int_0^{9.5} \int_0^{2\pi} \int_0^{5.5} (354.62r^2)(r)\,dr\,d\theta\,dz \\ &= 354.62 \left(\int_0^{5.5} r^3\,dr \right) \left(\int_0^{2\pi} d\theta \right) \left(\int_0^{9.5} dz \right) \\ &= 354.62 \left[\frac{r^4}{4} \right]_0^{5.5} [\theta]_0^{2\pi} [z]_0^{9.5} \\ &= 354.62 \left(\frac{14641}{64} \right) (2\pi)(9.5) \\ &= 1,540,798.1171875\pi \\ &\approx \boxed{4,842,364.3742} \end{aligned}$$

III.3 Matlab Implementation

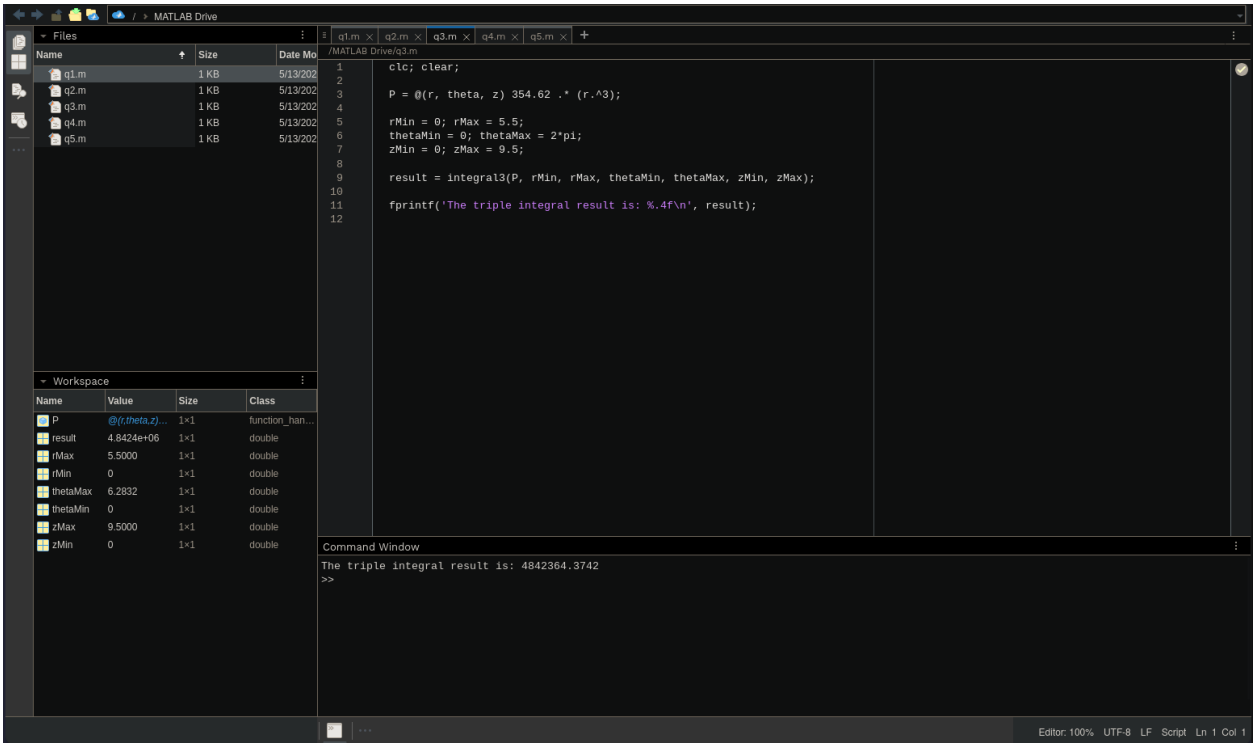

Figure III.1: Student 04 - Matlab Screenshot

III.4 Conclusion

The total tendon stress calculated by using cylindrical coordinates (Equation III.2) **matches** the MATLAB result (Figure III.1).

IV Musculoskeletal Force Output

IV.1 Problem Statement

In a multivariable musculoskeletal model, the **force output density function** for a limb system is:

$$P(x, y, z) = 901.17xyz$$

where x, y, z represent normalized inputs for muscle contraction, joint pressure, and bone displacement. A transformation to new coordinates is defined by:

$$x = \frac{u + 3v + w}{7}, \quad y = \frac{u - v + w}{7}, \quad z = \frac{u - v - 3w}{7}$$

Compute the **total musculoskeletal force output** over the region: $0 \leq u \leq 3$, $-1 \leq v \leq 6$, $-1 \leq w \leq 4$, using the **Jacobian transform**.

IV.2 Detailed Solution

Sol. Using **Jacobian transformation**:

$$
\begin{aligned}
J = \frac{\partial(x,y,z)}{\partial(u,v,w)} &= \det \begin{bmatrix} 1/7 & 3/7 & 1/7 \\ 1/7 & -1/7 & 1/7 \\ 1/7 & -1/7 & -3/7 \end{bmatrix} = \left(\frac{1}{7}\right)^3 \cdot \det \begin{bmatrix} 1 & 3 & 1 \\ 1 & -1 & 1 \\ 1 & -1 & -3 \end{bmatrix} \\
&= \frac{1}{343}\Big[1((-1)(-3)-(1)(-1)) - 3((1)(-3)-(1)(1)) + 1((1)(-1)-(-1)(1))\Big] \\
&= \frac{1}{343}\Big[1(3+1) - 3(-3-1) + 1(-1+1)\Big] \\
&= \frac{16}{343}
\end{aligned}
$$

$$
\Rightarrow dV = |J|\, du\, dv\, dw = \frac{16}{343}\, du\, dv\, dw
$$

$$F = \int_{-1}^{4}\int_{-1}^{6}\int_{0}^{3} 901.17\left(\frac{u+3v+w}{7}\right)\left(\frac{u-v+w}{7}\right) \cdot \left(\frac{u-v-3w}{7}\right)\frac{16}{343}\,du\,dv\,dw$$

Let $A = u - v + w$:

$$\begin{aligned}
F &= \frac{901.17\cdot 16}{7^3\cdot 343}\int_{-1}^{4}\int_{-1}^{6}\int_{0}^{3} A(A+4v)(A-4w)\,du\,dv\,dw \\
\frac{7^3\cdot 343}{901.17\cdot 16}\cdot F &= \int_{-1}^{4}\int_{-1}^{6}\int_{0}^{3}\left(A^3+4(v-w)A^2-16vwA\right)\,du\,dv\,dw \\
&= \int_{-1}^{4}\int_{-1}^{6}\left[\frac{A^4}{4}+\frac{4(v-w)A^3}{3}-8vwA^2\right]_{u=0}^{u=3}\,dv\,dw \\
&= \int_{-1}^{4}\int_{-1}^{6}\Big(9v^3+(21w-22.5)v^2 \\
&\qquad -(21w^2+27w-9)v-(9w^3+22.5w^2+9w-20.25)\Big)\,dv\,dw \\
&= \int_{-1}^{4}\left(-63w^3-525w^2+983.5w+1585.5\right)\,dw \\
&= \left[\frac{-63w^4}{4}-\frac{525w^3}{3}+\frac{983.5w^2}{2}+1585.5w\right]_{-1}^{4} \\
&= \left[\frac{-63w^4}{4}-175w^3+491.75w^2+1585.5w\right]_{-1}^{4} \\
&= (-1022)-(-934.5) \\
&= -87.5 \\
\Rightarrow F &= (-87.5)\cdot\left(\frac{901.17\cdot 16}{7^3\cdot 343}\right) \approx \boxed{-10.7237}
\end{aligned}$$

IV.3 Matlab Implementation

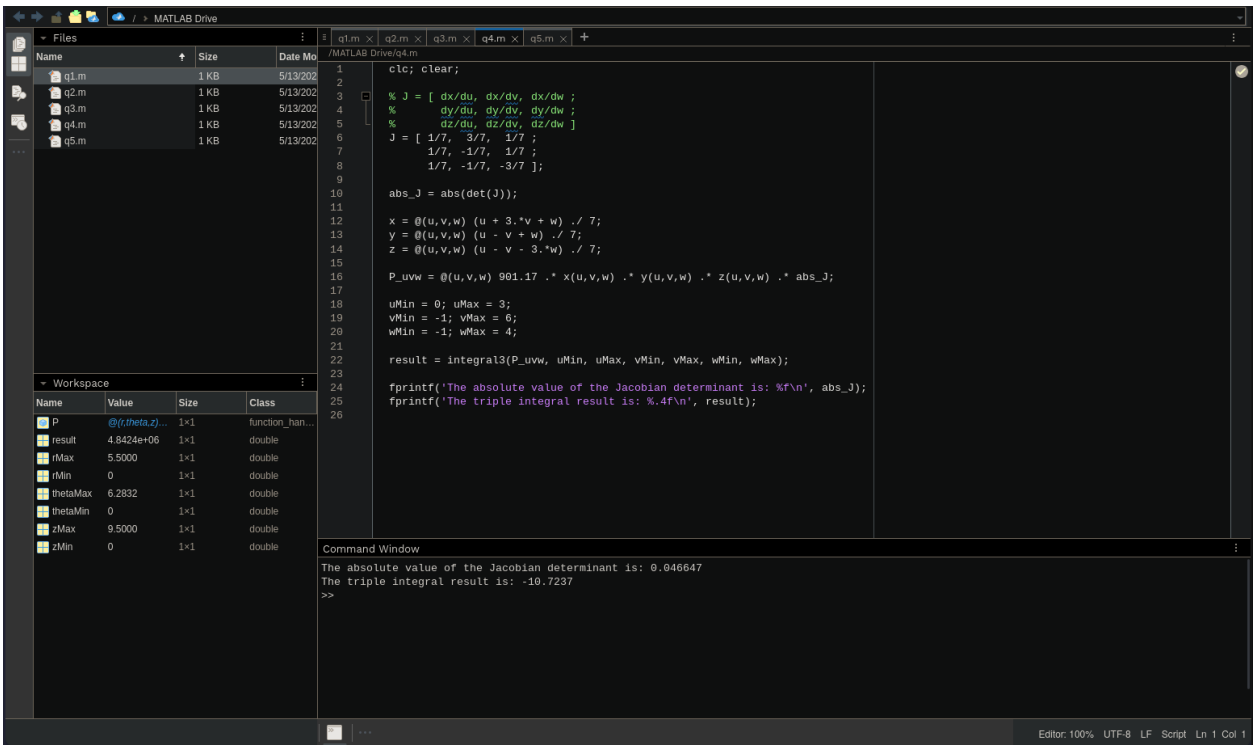

Figure IV.1: Student 04 - Matlab Screenshot

IV.4 Conclusion

After applying the Jacobian transformation, calculation for the force output (Equation IV.2) **matches** the MATLAB output (Figure IV.1).

V Force Cost Evaluation in a Tetrahedral Musculoskeletal System

V.1 Problem Statement

The following triple integral represents the **force cost function** in a bounded musculoskeletal system with quadratic variation along the vertical axis:

$$\iiint_E (480.92x + 592.03y + z^2)dV$$

where E is the solid tetrahedron bounded by: $x = 0.1$, $y = 0.3$, $z = 0.4$, $x + y + z = 5$. Evaluate the **total force cost** within the system.

V.2 Detailed Solution

Sol.

- **Inner bounds (z):** $x + y + z = 5 \Rightarrow z \in [0.4,\ 5 - x - y]$
- **Middle bounds (y):** $z_{\min} = 0.4 \Rightarrow y \in [0.3,\ 5 - 0.4 - x] \iff y \in [0.3,\ 4.6 - x]$
- **Outer bounds (x):** $y_{\min} = 0.3,\ z_{\min} = 0.4 \Rightarrow x \in [0.3,\ 5 - 0.3 - 0.4] \iff x \in [0.3,\ 4.3]$

$$
\begin{aligned}
C &= \iiint_E (480.92x + 592.03y + z^2)\, dV \\
&= \int_{0.1}^{4.3} \int_{0.3}^{4.6-x} \int_{0.4}^{5-x-y} (480.92x + 592.03y + z^2)\, dz\, dy\, dx \\
&= \int_{0.1}^{4.3} \int_{0.3}^{4.6-x} \left[(480.92x + 592.03y)z + \frac{z^3}{3} \right]_{0.4}^{5-x-y} dy\, dx \\
&= \int_{0.1}^{4.3} \int_{0.3}^{4.6-x} \Bigg((480.92x + 592.03y)(4.6 - x - y) \\
&\qquad + \frac{(5-x-y)^3}{3} - \frac{0.064}{3} \Bigg) dy\, dx \\
&= \int_{0.1}^{4.3} \Bigg[480.92x \left((4.6-x)y - \frac{y^2}{2} \right) + 592.03 \left(\frac{(4.6-x)y^2}{2} - \frac{y^3}{3} \right) \\
&\qquad - \frac{(5-x-y)^4}{12} - \frac{0.064y}{3} \Bigg]_{y=0.3}^{y=4.6-x} dx \\
&= \int_{0.1}^{4.3} \Bigg(240.46x(4.6-x)^2 + \frac{592.03}{6}(4.6-x)^3 - \frac{0.064(4.6-x)}{3} - \frac{0.4^4}{12} \\
&\qquad - 480.92x(1.335 - 0.3x) - 592.03(0.045(4.6-x) - 0.009) + \frac{(4.7-x)^4}{12} + 0.0064 \Bigg) dx \\
&= \Bigg[240.46 \left(\frac{x^4}{4} - \frac{9.2x^3}{3} + \frac{21.16x^2}{2} \right) - \frac{592.03(4.6-x)^4}{24} \\
&\qquad + \frac{0.04(4.6-x)^2}{6} - \frac{0.0256x}{12} - 480.92 \left(\frac{1.335x^2}{2} - \frac{0.3x^3}{3} \right) \\
&\qquad + \frac{26.64135(4.6-x)^2}{2} + 5.32827x - \frac{(4.7-x)^5}{60} + 0.0064x \Bigg]_{0.1}^{4.3} \\
&\approx \boxed{16,732.3117}
\end{aligned}
$$

V.3 Matlab Implementation

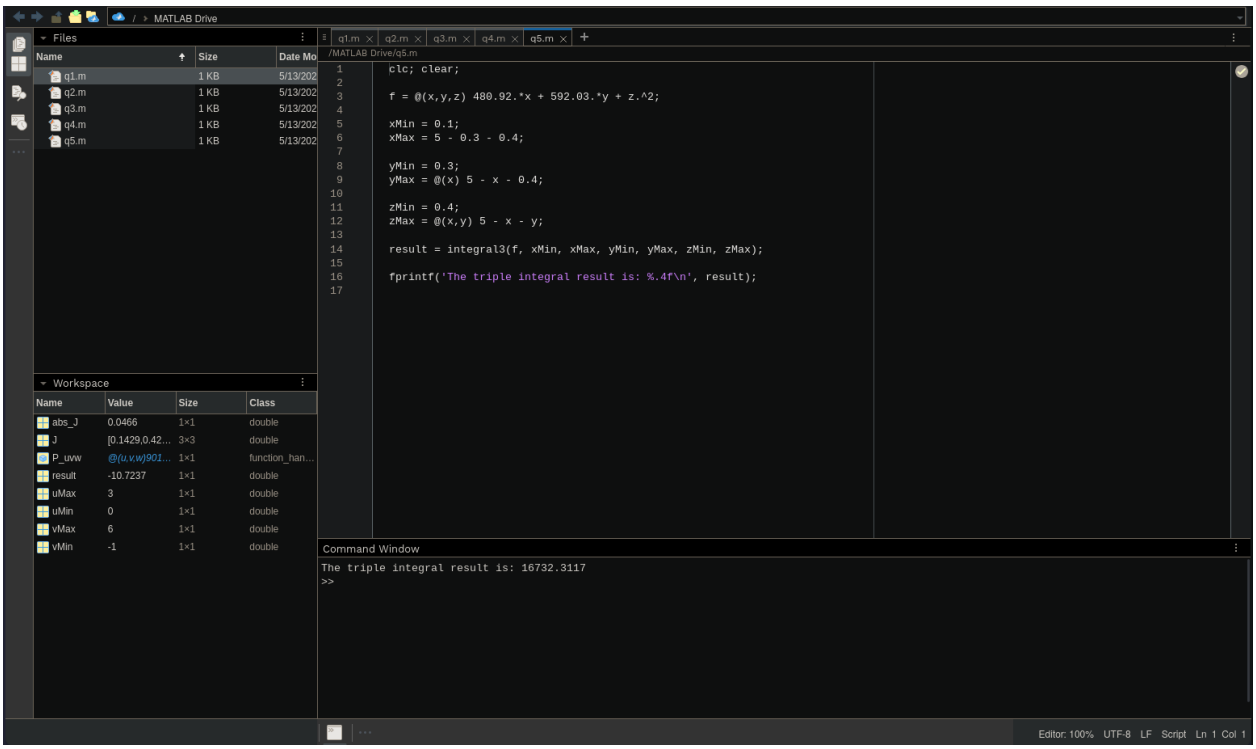

Figure V.1: Student 04 - Matlab Screenshot

V.4 Conclusion

The integration over the tetrahedral region (Equation V.2) results in the **same** value as the MATLAB script (Figure V.1).